Milestone 4 · Written analysis

Mahokshahvata (AI investment platform) · Google Cloud project `project-262a2026-879b-4b49-88e` · app image `v4` · October 2026

> *If our VM were deleted right now, what would break and what would survive?*

We would lose uptime, not data or configuration. Someone running two scripts brings the site back in about ten minutes.

What survives

Account and portfolio data is in Cloud SQL for PostgreSQL, a separate managed service on a private IP with daily backups and deletion protection. Profile photos, the market snapshot and sentiment history are in a private Cloud Storage bucket, and the database password and Anthropic API key are in Secret Manager. The app images (`v1` to `v4`) stay in Artifact Registry.

The static IP 34.122.164.209 is a separate reserved address, so the URL and Google sign-in's authorized domain keep working on a new VM. The firewall rules, the service account and its role bindings, and the budget alerts are project resources and survive as well. Sessions are rows in Cloud SQL, so users stay signed in.

What breaks

The website is down until someone builds a new VM. No load balancer, managed instance group or second VM can take over. Recovery is manual: someone with `gcloud` access to the project runs `./deploy/gcp.sh` and then `TAG=v4 ./deploy/docker_deploy.sh`, which takes about 10 minutes to create the VM, install Docker, pull the image and start the app. Neither script copies state or keys; the new VM runs as `mahokshahvata-vm` and reaches everything through IAM.

Anything held in memory is lost and then rebuilt on its own. That includes the login-throttle counters, so an attacker would get a fresh 5 attempts per email, and the caches for headlines, stock explanations and price charts. Re-classifying the headlines with Claude costs a few cents.

The autopilot can miss a day, because its scheduler runs inside the app. An outage during the 10:00 to 15:30 New York trading window skips that day's run. An outage over the close means that day's closing value isn't recorded, so the next daily-loss check compares against an older value.

Container logs are lost. Docker keeps them on the VM's disk and doesn't ship them to Cloud Logging (the role is granted, but no logging driver is configured), so the record of the minutes before the deletion would be gone. Switching Docker's log driver to `gcplogs` would fix this.

Zone risk, separate from the VM

Both halves run in a single zone: the VM in `us-central1-a` and Cloud SQL in `us-central1-c`, with no standby. An outage in either zone stops the site, though the data stays safe in Cloud SQL's 7 daily backups in Google's US multi-region. Cloud SQL's high-availability option, a standby in a second zone, would remove the database's zone risk at about twice its cost.